

Ogwi — Complete Functional & System Specification

Handover Document 2 of 4 · v2.1 (context edition) · August 2026 · Standalone and solely normative: no prior context required. *All rules identical to v2.0 (owner-ratified); this edition adds orientation and context passages only.*

You are here. Document 1 gave you the meaning — the person, the problem, the philosophy, and a plain-language tour of the machine. This document is where all of it becomes law. It is long because the product is decided in full, but it is built to be read in order: Part A walks the learner-visible world roughly as a learner meets it; a Bridge then retells the twelve underlying systems in plain language and traces one answer's journey through all of them; Part B specifies those systems in full, each opening with a short *in plain terms* orientation before its rules; Part C is the skeleton the behaviours hang on; Part D is the honest short list of what remains open.

How to read this. Part A specifies every product surface and feature. Part B specifies the twelve underlying systems — responsibilities, calculations, states, triggers, read/write rules, ownership, interfaces, edge cases, failure behaviour, configuration, analytics, privacy and QA acceptance criteria. Part C reproduces the normative architecture and governance (the canonical map, event model, configuration registry, publish points, invariants and sources of truth). Part D lists the small set of open decisions. Where a behaviour is not described in this document, it is not specified — do not invent it; raise it. **Configuration rule:** every numeric value explicitly assigned to a named Config document (LitreConfig, PhysicsConfig, SchedulerConfig, OggiConfig, ReadinessConfig, MasteryConfig) or marked (*config*) is team-tunable without code change under Part C's governance law; **every other numeric value is law.** Context passages — the Bridge, and every block beginning *In plain terms* — carry no independent authority: where a context sentence and a rule could ever be read differently, the rule wins.

Global vocabulary rules (apply to every surface): learner-facing copy never contains "streak", "failed", a bare "wrong" verdict, "overdue", "behind", "lost", "fading", or system vocabulary (rubric, key point, semantic, status, matched); "under" is banned as a judgement label — the mathematical phrase "under 20%" in the readiness climb frame is its sole permitted use; "not yet" is the only failure state; Flight copy is in Oggi's first person; "Helium" is the umbrella term, litres the unit; pending items use date phrasing ("Ready Thursday"), never countdowns; time estimates are rounded ("~3 min"), never timers.

PART A — SURFACES & FEATURES

Part A is the learner-visible world, ordered roughly as a learner meets it: the structure and the shell they move through (A1–A2) · the content they learn from and are tested by (A3–A5A) · the three active exercises (A6–A8) · the self-directed areas and their own progress picture (A9–A11) · Oggi, the sky, and the awards (A12–A14) · the people and money around the product (A15–A18) ·

the platform floor and the privacy summary (A19–A20). Each section states behaviour and copy obligations; the machinery underneath is cross-referenced into Part B, where it is specified in full.

A1. Content hierarchy

Qualification → Module → Topic → Lesson → Page. The Topic is the unit of the learning flow, sized at content-build time to **5–10 new knowledge items**; larger syllabus chunks split into sibling topics; genuinely indivisible objectives use a split-quiz flow and span sittings. Modules carry the awarding body's blueprint weights (each module's share of the exam); objectives carry published sub-weights where they exist, equal split otherwise.

A2. The immersive shell

Two worlds. The **Platform** (home, Practice, Progress, awards) uses normal navigation. The **Flow** (inside a Topic) strips all chrome to a minimal top strip — cross (exits via the exit popup) · topic name · a thin progress thread — and the **Dynamic Island**, the single adaptive hub. Three faces share one shell: the **Feed** (theory's building scroll), the **Canvas** (conversational activities: Teach Oggi, the Debrief; blurting uses the same canvas with Oggi absent), and the **Quiz frame**. **Exam mode is the strictest face:** the island slims to exam controls only — no Oggi, no litre counter, no listen, no hints, no Pomodoro.

The island's states: resting bar (Oggi avatar · contextual middle · progress ring + litre counter) · answer feedback · the progress sheet (topic contents with rounded time estimates) · the text/voice input transform for Canvas activities · navigation states at break points ("Continue?" / "Review first?" / "Not yet") · the learner-invoked, pausable **Pomodoro chip**. Desktop shows Oggi as a side panel; mobile is full-screen.

Topic Introduction: title, core idea, exam objectives, why it matters, the flow ahead with litre values and time estimates, and the reassurance line: progress is saved anytime, and Helium is never lost.

Exits. Lesson exit popup: *"Taking a quick break? Jump back in anytime — your progress is saved,"* with Oggi's *"I'm keeping every litre you earned"* (Keep Learning / Save and Exit). Exam exit popup: *"Your work is auto-saved, and the timer pauses with you"* (Keep Testing / Pause for Later / Grade My Answers, with one honest confirm listing unanswered questions).

The completion ceremony. Litres accumulate as **pending** during a sitting (display only); **every exit pumps** (B9). Topic completion plays the full ceremony: the checklist ticks everything covered (items from earlier sittings tick instantly as "already banked", never re-counted), a count-up of **this sitting's litres only**, then a **state-aware** closing animation — the single combined climb if Oggi is airborne or the sitting crosses liftoff; otherwise the takeoff gauge fills honestly ("20L in the tank — 80 to liftoff"). Mid-topic Save-and-Exit plays a light beat (count-up — payout, no ceremony). The economy is calibrated so the moment lands (B8): any ordinarily productive session that started grounded and completes its topic reaches liftoff; only a near-total-failure sitting falls short, and it lands with the takeoff gauge nearly full — one opener from the sky.

Flow-wide conventions: listen narration on all lesson and quiz types (never exam runs); speech-to-text beside every written answer; report buttons on lessons, quizzes and every AI

surface; a subtle "How was this?" lesson-feedback control; "i" tooltips explaining any new concept.

A3. Theory lessons and the authoring law

A continuous **building feed** — one idea per page, scrollable back through everything covered — carrying four callout blocks (Key Insight, Common Mistake, Example, Analogy), interactive HTML elements of any kind that aid the idea, tap-to-open glossary popups feeding a searchable student glossary, and a report-error button. **No interactive questions exist inside lessons** — rhetorical questions and hypophora keep the mind engaged in prose; checking happens in the quiz lesson type. Every lesson ends with a one-tap **confidence check** (Not yet / Almost / Got it); anything but Got it opens Oggi; the check feeds no scoring system.

The authoring law (build-time validation gates; a failing page cannot publish): one idea per page · **130–170 words per page, hard max 190**, embedded artifacts excluded, indivisible worked examples exempt · reading age 11–13 (Flesch–Kincaid grade 6–8) with exam-blueprint terminology always in its true form · the eleven explanation methods available to authors (plain definition, analogy, concrete example, use-case, step-by-step, contrast, visual description, in-a-sentence, build-up, misconception correction, exam-lens framing) · at least one example per important idea · career relevance woven in · connected flow without formulaic transitions · a per-page tooltip glossary.

A4. The in-flow quiz

Instant per-question feedback **with the explanation why**. **No in-the-moment retry:** the question returns in the **wrong-answer lap** at quiz end ("try the ones you got wrong again"), and again days later asked differently (B3) — rethinking at three time points instead of one re-guess. UI: litre counter, exam-objective tag per question, Oggi help, progress bar, question-type label, a **50/50 hint button** (multiple-choice only, one use per question, litres unaffected), listen audio, report. On any wrong answer the learner may enter the **clarification loop**: Oggi shows where the reasoning failed and can reteach with a fresh example. **Self-explanation prompts** ("why does that work?") fire only on newly learned or previously-wrong items, at most once per quiz, always skippable, never in exam mode. Answer types: multiple choice, multiple response, true/false, typed short answer, AI-marked long answer, sequencing, matching, fill-in-the-blanks, sorting, pick-an-image — recognition formats early, production formats as mastery grows (B3's ladder). Question stems span short and long written forms and may carry images. Flow quizzes are untimed by construction; slot-fill rules are in B5.

A5. Exam mode and assessment runs

Formal variant: select + Submit-to-confirm, a timer that pauses with the learner, Back/Next with a question navigation panel, "Question 4 of 30", flaggable questions — with everything gamified stripped (no litre counter, hints, Oggi, explanations, instant feedback, listen). **Exam Simulation:** one entry generating a fresh randomised paper each time (question count and allotted time shown). **Mini-mocks:** 10–20 questions in the same bucket. The **first full mock arrives by invitation** once mastery reaches a threshold (*config; see Open Decisions*), framed as a gap-finder; after any rough mock, Oggi opens with a normalising script plus the two highest-

leverage fixes. **Mock scores are private forever.** Litres accrue silently and pay at the feedback screen with completion premiums (B8).

A5A. Feedback and answer review

Every attempt follows **goal** → **what happened** → **next step**. A bare "wrong" verdict is prohibited; feedback addresses effort and strategy, never the learner as a person; "not yet" is the only failure state.

Three context-specific result screens: **In-flow quizzes** end with a light screen — score, one-line strength, one-line weakness, and a statement that missed items have been queued (every question already gave instant feedback and the wrong-answer lap ran). **Standalone practice quizzes** end with the full screen — score; Review Answers; Retake and Continue; strengths; weaknesses; one suggested next step at module, topic or lesson level; and the personalised-quiz notice. **Time is never displayed unless the learner chose a timer.** **Exam-format runs** add pass/fail against the qualification pass mark and time used versus time allotted.

Review Answers shows, for every question: the result marker · the learner's answer · the correct answer · an explanation in the form "*[answer] is correct because...*" · an Oggi talk-through action wherever Oggi is permitted · and the source topic/lesson/objective it came from.

A6. Blurting (surface)

Setup: the learner picks the timer (3 minutes / 5 minutes / unlimited; pausable), then types or speaks everything they remember; the canvas is memory-only (no Oggi, no AI help). The **hint button** is unlimited: each press reveals one pre-authored cue question pointing at an untouched area while containing none of its content; hint use is never penalised or mentioned. The results screen shows exactly three numbers — **Coverage, Accuracy, Critical gaps** (each gap named in ten plain words or fewer) — and three plain lists: what you got right, what you missed, what you got wrong. Two follow-ups: **Correct Mistakes** (the blurt annotated green/amber/red; jump-to-mistake iterates the red items only, each with what was off and the correct idea beside a rewrite box; every correction skippable; finishing or skipping all red items ends the flow) and **Continue**. Scoring mechanics: B6.

A7. Teach Oggi (surface)

Oggi plays a genuinely confused student; the learner explains the topic in their own words. Per-turn **chips** carry all grading (Great / Good / Inaccurate / Mistake); Oggi's voice never says "wrong" — flawed explanations earn innocent follow-up questions aimed at the flaw. A **learning-objectives panel** ticks as objectives are covered; a **knowledge meter** fills toward **Confident**, which triggers the close: Oggi's thank-you ("hey — I'm feeling confident about subnetting now, thank you!") and the identity beat ("You just taught subnetting. Teachers know their stuff."). **Teaching Notes** are available throughout: a static crib sheet showing the whole topic's plan — every point's plain name and its cue question, never any explanation text. **End Session** is always available and pays out what was taught, with a report (full on natural close: strengths quoting the learner's best formulations, improvements in plain words, "not covered yet" without blame; light on early end), transcript attached. Mechanics: B7.

A8. Oggi the Debrief (surface)

A 3–5 minute consolidation conversation at each module's end, chat or voice: the learner's two-sentence summary — two or three probing questions on what the summary skipped — one connection to an earlier module — the learner names what feels shakiest (which visibly shapes tomorrow's session) — a preview of what's next. Ending early is always allowed. Mechanics: B11.

A9. The Practice page

1. **Jump back in** — paused quizzes and exams with questions remaining; deletable.
2. **Exam Simulation** — the single randomised practice-exam entry.
3. **Create a test** — name it; draw questions by modules, topics, exam objectives, or **the wrong-answer pool** (definition in B3; shown with a live count; empty state: *"Nothing here — you've cleared your misses"*); question count in tens; quiz mode (instant explanations, lives, hints, timer, hide-answers, allow-Oggi — each configurable) or exam mode (pass mark, fixed time, no aids).
4. **Recommended** — where the engines deliver, each quiz labelled for what it is: *"Fixing gaps in: [Module]" / "Keeping fresh: [Module]"*; pending items date-phrased: *"Ready Thursday — that's when reviewing it helps most."*
5. **Activities** — learner-chosen exercises: **Blurt** any covered topic · **Teach Oggi** any covered topic · **Debrief** any completed module. Mechanically identical to their in-flow versions (same scoring, hints, writes — B6/B7/B11); topics with no covered content don't list; **no pairing restrictions apply to learner-initiated activities** (blurting then teaching the same topic in one day is a legitimate choice). Housed separately from Create a Test because a test is questions asked of you and these are things you do.
6. **Completed** — every past test framed progress-forward: latest score leads; previous attempts sit one tap deeper as date-labelled entries ("12 Mar · 54%") with improvements highlighted (" +17 since last time"); never numbered attempts; retake and delete.

A10. The Progress page

Part 1: programme completion % and the expected finish date — always *"at your current pace: ~14 March"*; freezes (never inflates) after 14+ fully quiet days, with *"picks back up when you do."* **Part 2 — "This week so far":** a 7-day chart of litres earned per day, weekly summary (litres · sessions · topics), skimmable previous weeks; empty days simply empty. **Part 3:** the mastery trend; per-module mastery against the pass-mark line with **mastered** / **"due for a refresh"** states; the **Biggest Opportunity** with leverage framing ("it's 18% of the exam"); and the readiness odds with certainty, next action, and — pre-unlock — the three-item checklist ticking toward the first score.

A11. Onboarding and selection

Browse careers — see required qualifications — pick one — begin. No placement test, no target-setting step. Jargon-free copy with the primer and glossary; the ten-second Flight explainer at first liftoff; the first-session standard: *"I'm in good hands."* Early re-routing is free — *"new destination, same balloon"*: flight records and helium persist; mastery is per-qualification. The trial ends when the first session completes.

A12. Oggi surfaces and modes

Presence rule: support-Oggi appears on theory lessons, in-flow quizzes and quiz-mode practice; **absent from exam runs** (unaided), **absent from blurts** (memory only), **absent as a helper in Teach** (he's the student; the Notes are the help). A general chat exists outside the flow. **Four modes:** contextual quick prompts (curated per moment, at most four, always a write-your-own; e.g. theory: explain simply / quiz me / key concepts / how would the exam ask this; wrong answer: where did I go wrong / reteach with a new example / give me a similar question), typing, speaking, and a call with live transcript. **The greeting:** one line maximum, input free from the first moment; at most one *invitation* per session when something genuinely storied exists ("Want to pick back up where subnetting got tricky — or something else?"), never a required question, never re-asked. **Furniture:** stop button, "Thinking" state, listen-with-word-fade, Oggi's headshot, "Oggi can make mistakes" notice, helpfulness thumbs, report button, in-chat Quiz Me (casual; feeds no scoring). Chats look fresh (no visible history; "your chats are private") while the learner model persists (B11). The **"what Oggi knows about you" panel** shows his notes verbatim, all editable and deletable. Oggi never contacts anyone outside the app except the Flight System's single push, in his voice.

A13. Flight display and notifications

Homepage: Oggi bobbing at his **exact** altitude against the city, with Flight time ("Day 9 aloft"), Altitude, Helium, and Longest Flight; grounded, a takeoff bar filling 0–100L with the exact figure. The post-lesson climb is the only motion in the system — one combined, state-aware climb per sitting; band crossings earn an extra beat *within* that single animation. Descent is never animated or narrated; below 100ft one line appears: *"I'm drifting low. One lesson tops me up."* **Exactly one push exists:** *"I'm running low on helium. One lesson tops me up"* — trigger and caps in B9. Touchdown is never pushed; it is shown warmly on next open as a finished story. The Flight Log lists completed flights ("Flight 7 — 12 days aloft · peak 460 ft · 1,800L pumped"); the records board: Longest Flight, Highest Altitude, Biggest Day, lifetime litres, lifetime days aloft. A "most flights" figure is never computed or shown. Light mode renders day, dark mode night, with seasonal effects. Above the 1000ft cap, the Orbit display state applies (B9).

A14. Awards and cosmetics

Altitude firsts: First Lift (100) · Above the Rooftops (200) · High-rise View (300) · Among the Towers (400) · **Breaking the Skyline (500)** · Among the Cranes (600) · The Spires (700) · Broadcast Heights (800) · The Warning Lights (900) · **The Ceiling (1000)** · **Orbit** — visible from day one, locked, its requirement shown only as **"??? ft"** (the award is the clue that something exists above the top of the sky), granted on the first cap-overflow. **Flight lengths:** 3 · 5 · 7 (First Week Aloft) · 10 · 14 (Fortnight Flyer) · 20 · 30 (A Month in the Air) · 50 · 75 · 100 (Century Flight) · 150 · 200 · 365 (A Year Aloft). **Big days** (litres in a local calendar day): Double Day 200 · Triple Day 300 · Big Pump 500 · The Thousand 1,000 (pumped from the ground, it lands exactly at the Ceiling). **Lifetime:** litres 1k/5k/10k/25k/50k/100k; days aloft 30/100/250/365/1,000. **Sustained (the elite set; semantics in B12):** Tower Residency (400ft × 7 days) · Skyline Week (500 × 7) · Skyline Fortnight (500 × 14) · Spire Week (700 × 7) · Spire Day (700 held continuously for 24h) · Cruise Control (mean 300ft+ across a 30-day flight). **Rhythm:** Daily Lift ×7/×30 (a

top-up on 7 / 30 consecutive days) · Full Days ×7 (100L+ on 7 consecutive days) · Commuter's Climb (5 consecutive weekdays) · Weekend Lift (200L+ in a weekend) · Early Air (10 top-ups before 8am). **Comebacks (one-time)**: Straight Back Up (relaunch within 24h) · The Long Way Back (relaunch after 30+ days) · New Wind (beat your Longest Flight). **Course tie-ins**: Module Aloft (a module completed within one flight) · Mock at Altitude (mock completed above 300ft) · Skyline Scholar (above 500ft). **Firsts**: First Liftoff · First Band Crossed · Flight One · Record Setter.

Catalogue law: every axis carries one very difficult capstone; all awards are one-time; awards celebrate and never nag — no countdowns, warnings or pushes; nothing rewards touching down; sustained beats spiked; day-counting uses the learner's local calendar while physics stays rolling; **the fairness test**: every award must be earnable by a parent who lost a week and came back. Awards are earned at the exactly-computed crossing moment (even when discovered later, the record shows the honest earned-on date), surfaced at the next natural moment, and recapped in the touchdown story. **Cosmetics**: completing a catalogue unlocks a city theme; each award unlocks an avatar; Oggi himself keeps the brand look; seasonal challenges offer limited-run cosmetics, never physics.

A15. The Wingman system

Accountability through chosen people. **Laws: private by default; every disclosure chosen, revocable, previewable; encouragement never buys altitude; no shame surfaces anywhere.** Multiple Wingmen from day one, framed as one ("pick your Wingman", always an "add another"), each independent — own link, own digest, no shared feed. The invite prompt fires after the first completed session post-subscription; declining is one tap; offered at most three times ever, with the public Flight Tracker presented as the sibling choice. Invite flow: pick the person — **preview exactly what they'll see** — send a prefilled editable message by any channel. **Disclosure menu (learner-toggled)**: always — first name/photo, qualification, live flight status, next milestone; on by default — modules-mastered bar, Longest Flight, comeback count; off by default — overall mastery %, readiness odds, studied-days chip, total study time; never on the menu — attempt counts, exact last-active timestamps, activity calendars.

Capability links: a single-purpose invite link per Wingman — acceptance captures only an email (consent runs both ways) — a personal snapshot link (long unguessable token; any browser, signed out, indefinitely; digests deep-link to it so a lost link self-heals); HTTPS-only, noindex, referrer-stripped, rate-limited; two-tap remove/swap/mute with a kind sunset page; token rotation on leaks; Wingman step-down (a Wingman can remove themselves); deletion on request. **Weekly digest** (optional both sides): learner-picked day, three lines maximum, encouragement buttons, one soft footer ("Curious what this is? Start your own flight"); quiet weeks get one honest sentence and an invitation to encourage — never a zero, never a cadence change. **Encouragement** delivers instantly (chips or short filtered notes — a banner and a cosmetic flourish in the sky) and pays zero litres. **The Flight Tracker** (public, opt-in): alias mode available from day one; user-approved milestone cards only; anonymous public encouragement chips; no digests, likes or follower counts; share prompts only at success moments. **Instrumentation (all content-free, aggregate)**: invite-prompt funnel (shown — sent — accepted) · digest delivery and open rates · encouragement events per digest (counts only, never content) · snapshot-link visit counts · disclosure-toggle usage rates · mute/remove/swap/step-down events · Tracker opt-ins and milestone-card approvals · quiet-

week digest sends. Law: nothing per-Wingman identifiable beyond counts; no message content ever enters analytics.

A16. Arrival Story and referrals

Post-pass, the system drafts an editable **Arrival Story** card — *"After 142 days, 61 flights and 3 comebacks — [Name] has arrived: [QUAL], passed first time"* — with journey stats and the skyline. **Sharing it publicly earns the referral month(s)** — reward attached to proof. The flow carries a story-consent opt-in feeding the marketing library. Referrals are give-get: the friend receives an extended trial; the referrer's free month lands after the friend's first paid month; capped at 12/year; automated fraud checks with human spot audits; automatic disclosure tags; links live in the Arrival Story, Tracker pages, and a quiet settings surface.

A17. Money surfaces

Trial: no card; ends at first completed session. Subscription \$39/month. Cancel: two screens, no guilt copy. **Hold my balloon:** billing pauses; progress, mastery, flight records and Wingman connections persist; resume restores everything. The support reply-time promise is published on pricing.

A18. Support and operations

A small, named, human, **reactive** support desk — platform issues and light qualification questions only. No proactive outreach; no mentors, coaches, tutors, cohorts or scheduled live sessions, ever.

A19. Platform and technology

Mobile-first web plus native iOS and Android; a gentle app-store nudge only after the habit forms. Offline lesson packs pre-fetch on wifi and sync by **true timestamps** (safe by construction — see B9's correction rule). Server-authoritative rolling time throughout; device clocks are never an input. Speech-to-text on every written answer; listen narration per A2's rules.

A20. Privacy and safety summary

Chosen disclosure only — nothing about a learner reaches another human without their deliberate act. Oggi chats are private; support transcripts are retained short-term for the report button and QA only and are never mined into future context. Sensitive personal content is structurally unstorable in Oggi's memory (B11). Mock scores, mastery and readiness are private by default. Attempt counts are never displayed anywhere. Content flagged as dangerous or unusable follows the urgent pause route (B10). Oggi's capability boundary (B11) keeps him out of personal, emotional, medical and legal territory.

THE BRIDGE — THE MACHINE IN PLAIN LANGUAGE

Context only, never authority: this section exists so Part B reads as a story you already know. Where any sentence here could be read differently from a rule below, the rule wins.

How the twelve fit together. One versioned **content graph** (B10) is the single source of truth — every module, topic, objective, marking point, question and lesson page, human-approved before it can serve. Everything else is machinery reading that graph. The shared heartbeat is the set of **per-item memory states** (B4): every fact the learner has ever correctly retrieved carries Difficulty, Stability and a live recall probability. **Mastery** (B1) aggregates those states the way the exam weighs them and publishes a kind-but-honest display. **The adaptive engine** (B3) watches every miss and runs the repair loop. **Composition** (B5) assembles each session — the opener, the topic flow, the activity slot — so the learner never plans. **Blurting** (B6) and **Teach Oggi** (B7) mark free recall and free explanation against one shared rubric drawn from the graph. **The economy** (B8) prices every learning act in litres; **flight physics** (B9) turns litres into altitude with a pure function and an append-only history. **Readiness** (B2) converts the live mastery layer plus mock evidence into calibrated odds of passing, and quietly tells the spaced engine how long memory must last. **Oggi's systems** (B11) govern what he's given, remembers and may say on every surface. **B12 and Part C** close the loop: award semantics, the canonical map, and the invariants that block release.

The journey of one answer. 7:42pm; Susie, mid-quiz, answers a subnetting question — wrong. The rendering resolves to its knowledge item; one **Again** event writes: the spaced engine shrinks the item's Stability and its recall probability drops (B4); the adaptive engine flags the item into remediation with source and timestamp (B3); the economy pays 0L — litres never subtract (B8); the feedback law shows the goal, what happened, and why the right answer is right, and the question is queued for the end-of-quiz wrong-answer lap (A4, A5A). Live mastery dips by that one item's weight; the *displayed* score doesn't move mid-activity — at session end the publish points fire (Part C4): mastery publishes (easing any decline on its 7-day half-life), Biggest Opportunity and the odds recompute on the live layer only (B1, B2). Days later the item returns inside "*Fixing gaps in: Networking*" — batched with its module's other repairs, padded with confusable neighbours Susie already holds, at the right rung of the format ladder (B3). Two correct answers, on two different renderings, on two different days, and it exits repair back to ordinary spaced review; each of those correct-on-due answers pays Class B, the economy's top rate (B8). Meanwhile the session's litres pumped at the completion beat, Oggi climbed the exact difference (B9), and the odds of passing edged up (B2). One wrong tap; eleven systems did their jobs.

One learner-day, system by system. Open — the greet, at most an invitation (B11) — the opener: 4–8 due and repair items, remediation-first (B4/B3) — the topic flow per its build-time template (B5) — theory pages at 1L each (A3/B8) — the in-flow quiz, uncovered new items first (B5/A4) — the activity slot resolves to a blurt (B5–B6) — the completion ceremony, the grounded-liftoff guarantee if it applies (A2/B8) — the pump, the climb, the honest altitude (B9) — publish points — Recommended refreshed for tomorrow (Part C4).

The dependency spine (the intuition behind Document 3 §8's build order): graph — memory states — mastery — the two engines — composition — the rubric activities — the economy — the flight — readiness — Oggi — the invariants over all of it.

PART B — THE TWELVE SYSTEMS

B1. Mastery Scoring

In plain terms — the honest scoreboard. Every fact the learner can actually retrieve carries a probability; modules average those exactly as the exam weighs them; the display is kind about the timing of bad news but never lies upward into anything that predicts. Builds on: the content graph (B10) and the memory states (B4). Feeds: readiness (B2), the Progress page, Biggest Opportunity.

Responsibility. One honest, kind number per module answering "how much of this do I hold?", plus the Biggest Opportunity pointer. Owner of: the mapping table, published mastery values, mastered-state transitions.

The atom. Every knowledge item carries an **FSRS memory state**. FSRS (Free Spaced Repetition Scheduler) is the published open-source scheduling algorithm this product adopts wholesale: its Difficulty/Stability/Retrievability model, its retrievability formula, its state-update equations on review, and its published default parameter set are the normative definitions wherever this specification says "scheduler." The state is Difficulty, Stability, and derived **Retrievability R** — the probability of correct recall right now, 0–1. **Item mastery = current R**. An item enters scoring at its **first correct retrieval**; never-retrieved items score exactly 0 regardless of reading.

The mapping table (single source of truth). item $\leftrightarrow$ objective $\leftrightarrow$ module $\leftrightarrow$ blueprint weight. Objective score = unweighted mean of its items' R. Module score = mean of objective scores, sub-weighted where the awarding body publishes sub-weights, equal split otherwise. Programme mastery = $\Sigma(\text{module score} \times \text{blueprint share})$. "72% mastery" therefore means "predicted to hold 72% of the exam, as the exam counts it." Mastery, readiness, both engines, quiz tagging and Biggest Opportunity all read this one table.

Evidence rules. Every graded question anywhere writes one review event on its item: **correct — Good; incorrect — Again**. Blurting and Teach Oggi write at objective level at their single scoring moment: content correctly recalled/taught — Good per mapped item; content clearly wrong — Again on the specific items; content merely missed — **no grade — a due-date pull-forward only**. The Debrief writes on the same terms at its scoring moment, under the ambiguity law: cleanly matched points — Good; clearly wrong assertions — Again; everything merely unaddressed — nothing (B11). Nothing else moves mastery: the in-chat Quiz Me and the lesson confidence check write nothing.

One memory, many faces. Each item owns multiple **renderings** (base question, ≥ 2 variants, format-ladder versions: multiple-choice / cued / typed). Any rendering's grade writes to the item's single state. Never serve the same rendering twice consecutively for an item; each rendering logs last-served.

Decay. $R(t) = (1 + 0.2346t/S)^{-0.5}$ — a decelerating power curve calibrated so $R = 90\%$ when elapsed time equals Stability; it never reaches zero, and no artificial softening or cap is applied. Reference points ($S \approx 30$ days): $\sim 90\%$ after an untouched month, $\sim 72\%$ after four, $\sim 51\%$ after a year. Every successful review multiplies Stability, so well-known material decays slower and maintenance cost shrinks over time.

Publishing (the display layer). Publish points: session end, Practice-run end, one daily rollover during inactivity — never mid-activity. If live $\geq$ displayed, the display **jumps to live** (gains land instantly, same sitting). If live $<$ displayed, the display converges with a **7-day half-life of the gap** (*config*): $\text{displayed} += (1 - 0.5^{(\Delta t/7d)}) \times (\text{live} - \text{displayed})$, closed-form on rolling time. There is no delay before movement — the half-life is a speed, not a wait. Invariant: displayed only ever sits at-or-above live, bounded and temporary. Only published values appear anywhere; raw instantaneous values are never surfaced. Learner-facing mastery values display as whole percentages; full precision is retained internally.

The honesty split. The displayed value drives labels, the trend and Biggest Opportunity. **Both engines and the readiness computation read live values only.** Acceptance rule: no prediction surfaced to a learner is ever computed from a smoothed input.

States. A module is **mastered** when its displayed score $\geq$ **the qualification's pass mark** (the single bar product-wide; no per-learner target exists). Decay below the line relaxes the state to **"due for a refresh"** — never "lost", never red — and it flips back on recrossing. Entering the refresh state requires no learner action: the spaced engine is already queueing exactly those items.

Biggest Opportunity. $\text{drag} = \text{blueprint weight} \times \max(0, \text{pass mark} - \text{displayed mastery})$; the maximum wins; ties break to the heavier weight; recomputed at session/run end, stable between; surfaced with impact framing. Worked check: 20%-weight module at 55 (bar 75) drags 4.0; 5%-weight module at 30 drags 2.25 — the first wins despite the less scary score.

The downside ledger (ships as an acceptance-test suite, kept green). Only two things can lower mastery: an Again on an item with a prior correct retrieval, and time decay. These never can: correct answers of any kind; first-attempt failures on unentered items; reading or not reading; missed days (no zero-events are written for absence); blurt-missed content; confidence checks; Quiz Me; hint usage; ending early.

Recovery guarantee. One correct answer on any previously-known item restores $R \approx 100\%$ **and** grows Stability past its prior value — every comeback slows the next decay. Theory is reserved only when refresh attempts fail (B3's recap-then-retry).

Failure behaviour. If publishing is interrupted, the next publish point recomputes from live state — the display layer is derivable, never authoritative. **Analytics:** published values, state transitions, Biggest Opportunity history. **QA acceptance:** the downside ledger; the at-or-above-live invariant; the jump-on-gain behaviour; the pass-mark threshold on both states and drag.

B2. Readiness & the Completion Forecast

In plain terms — the answer to "am I ready?", stated as checkable odds rather than a vibe: a projected exam score, calibrated against how the learner actually performs under exam conditions, converted into a probability of passing with its certainty said in plain words. Builds on: B1's live layer and the mock-run log. Feeds: the Progress page, and — quietly — the spaced engine's horizon (how long memory must last).

Responsibility. The odds of passing (with certainty and a next action), the one-time readiness celebration, the completion forecast, and the spacing horizon. Owner of: published odds, certainty band, next action, forecast, horizon.

Internal projection (μ). Every item's live R projected forward **14 days** (*config*) on its own curve, aggregated through the mapping table → a predicted exam score. Rationale: booking-to-sitting realistically takes ~two weeks, and Ogwi never knows exam dates; predicting "right now" would flatter.

Exam-condition calibration. For each completed exam-format run (Exam Simulation, mini-mock, custom exam mode): ratio = achieved score ÷ the projection at that moment. Calibrated μ = projection × the recency-weighted mean ratio, **clamped to [0.7, 1.1]** (*config*). No runs → ratio 1.0 (and certainty stays Early).

Uncertainty (σ). Composed from: untouched-coverage share (dominant early) · exam-format evidence count and recency · spread between run ratios · residual projection variance — with a floor (*config*) so precision is never faked. Thin evidence widens σ , pulling extreme odds toward the middle automatically.

The score. $P(\text{pass}) = \Phi((\mu - \text{pass mark})/\sigma)$, shown as a whole percentage, recomputed only at publish points, **live-layer inputs only**. Copy frame: *"Odds of passing if you sat it soon: 82%."*

Certainty bands (always stated with their basis in plain words): Solid = weighted coverage ≥70% and ≥2 exam-format runs in 30 days · Fair = ≥40% and ≥1 run in 45 days · Early = anything less (*all config*). **Weighted coverage, defined:** the blueprint-weighted share of knowledge items with at least one retrieval event — each item inherits its objective/module blueprint contribution through the mapping table; untouched items contribute zero. The same value feeds σ , the certainty bands and qualityRatio.

First-score unlock. The readiness surface begins as a visible three-item checklist, each ticking as earned: complete your first topic · answer questions across at least two modules · complete one mini-mock (10–20 questions, ~10 minutes). All achievable in week one by choice; because a run is required, every score ever shown is calibrated from birth. Reveal copy: *"Here's your first readiness score — it sharpens with everything you do."*

Display rules. Below **20% odds** (*config*) the figure is withheld for the climb frame: *"under 20% — right on track at 18% coverage; this number exists to climb."* From 20% up the number always shows with its stage anchor: qualityRatio = $\mu \div$ weighted coverage, compared to the pass mark — *"what you've covered, you're holding above pass level — keep going and the odds follow."* "Behind" and "failing" are banned, and "under" is banned as a judgement label — the mathematical "under 20%" of the climb frame is its sole permitted use; any gap renders only as the next action.

Next action. Simulate the odds-delta of three candidates — (a) refresh the largest decayed pool; (b) a session on the Biggest Opportunity; (c) a mini-mock when exam evidence is missing or stale (certainty-band gains count as improvement). Highest wins; ties prefer (c) then (a); output one plain line ("One mini-mock would make this number solid").

Celebration. Fires once, on a **rising crossing of 80% odds with certainty Fair or better** (*config*): "You'd very likely pass if you sat it soon." Later dips trigger nothing — no alarm, no retraction.

The forecast. Expected finish = today + remaining flow items ÷ paceEW, where paceEW is items/day exponentially weighted over the trailing 28 days with a 14-day half-life (*config*). Recomputed at publish points; always "at your current pace"; after 14+ fully quiet days it **freezes** at its last value.

The horizon (spaced-review scheduling only — content is never scheduled and no deadline exists). horizon = clamp(forecast – today, **floor 45 days, cap 12 months**; default **3 months** before pace data exists) (*config*). Read lazily at each per-item interval assignment; **a moved forecast never mass-reschedules existing due dates** — only future assignments use the new horizon. The floor prevents review pile-ups near completion; the cap keeps slow-pace horizons sane.

Failure behaviour. Missing pace data – default horizon; missing run data – ratio 1.0 + Early cap. **Analytics:** odds/certainty history, celebration event, calibration pairs (odds at sitting vs self-reported outcome) — the readiness-accuracy KPI. **QA acceptance:** live-only inputs; the 20% withhold; single-fire celebration; the never-reschedule guarantee; banned vocabulary absent.

B3. The Adaptive Engine

In plain terms — the repair shop. Every miss, from any surface, becomes a comeback that must be proven twice — asked a different way, on different days — with the question format stepping down toward recognition only when needed and climbing back before release. Builds on: grade events from everywhere. Feeds: opener ranking, "Fixing gaps" quizzes in Recommended, and the custom builder's wrong-answer pool.

Responsibility. Detect and repair weak spots. Owner of: the remediation bank, rung states, the gap-quiz queue, the wrong-answer pool definition.

Entry. Any Again event from any graded surface, plus objective-level "clearly wrong" writes from blurring, Teach Oggi and the Debrief, places the item in **remediation** (flagged with source and timestamp).

Exit. **Two correct retrievals, on two different renderings, on two different days.** Exit clears the flag and returns the item to the spaced engine alone.

Rendering ladder. First remediation serve: same format rung, different rendering (the same knowledge asked a different way). A further failure drops one rung toward recognition (typed – cued – multiple-choice; floor = multiple-choice); each success climbs one rung back, so items that entered at production leave being produced. Rung state persists on the item.

Escalation. Two failures at the multiple-choice floor – **recap-then-retry**: the item's linked one-idea lesson chunk surfaces as a recap card immediately before a fresh rendering. Each such event logs a **concept-breakdown signal** for content QA.

Batching. Per module: a "*Fixing gaps in: [Module]*" quiz generates at **≥5 remediation items or 7 days from the oldest unbatched signal**, whichever first; ready-date **2–4 days** out, load-spread across the week, date-phrased. Sub-threshold remainders ride the next session opener.

Gap-quiz composition. Remediation items (at their correct rungs) $\leq \sim 70\%$ of the quiz, padded with **held confusable neighbours** (items at $R \geq 0.8$ linked by a confusable edge — B10) so repair doubles as discrimination practice and the success rate stays humane. Length 8–15; loads above 15 split across days; interleaved ordering; standard quiz mechanics.

The wrong-answer pool (Create-a-test source). Items currently in remediation **plus items that exited within the last 30 days** (the retest window proving the fix held).

Caps and priority. Remediation $\leq 50\%$ of any opener; ≤ 1 gap quiz surfaced per day; priority = module drag, then oldest signal; **nothing ever expires**. A struggling learner's experience stays "a few things to fix today", never "your 60 failures."

Inventory contract (with B10). Every item ships with ≥ 3 renderings plus ladder versions; never the same rendering twice consecutively; prefer least-recently-served; if all renderings served within 14 days, serve the least-recent anyway and log an **inventory-exhaustion event** feeding the variant-generation queue.

Interface (nothing else reads the bank). Three exports, refreshed at publish points: per-item flags with rung state ($\rightarrow$ opener ranking, wrong-answer pool) · per-module **weakness share** (blueprint-sub-weighted fraction of entered items in remediation $\rightarrow$ session composition's in-surface weighting only; it never alters Biggest Opportunity) · the batch queue with ready-dates ($\rightarrow$ Recommended).

QA acceptance: the two-two-two exit; ladder monotonicity; the $\leq 70\% / \leq 50\% / \leq 1$ -per-day caps; nothing-expires; single-serve dual-write (B4).

B4. The Spaced Engine

In plain terms — the memory keeper, and the heartbeat of the whole platform: its per-item states are what mastery aggregates and readiness projects. It brings material back just before it would slip, never surfaces a backlog, and never punishes lateness — a late successful review strengthens memory more, so the comeback itself is the cure. Feeds: everything.

Responsibility. Keep memory: decide what is due and deliver review. Owner of: per-item scheduler states, due computation, the "Keeping fresh" queue, opener composition.

Due. An item is due when projected $R \leq 0.90$ (desired retention — a global constant, never a learner setting).

The opener. Every session begins with **4–8 items** (scaling with load; ~ 12 – 15 seconds each): overdue remediation items first ($\leq 50\%$), then due items by R ascending; an item answered correctly today never reappears in today's opener. **Never empty:** with nothing due, the 3 lowest- R entered items serve — sessions always begin with production, and early retrieval grades normally. **First-session exception:** when no item has yet entered the scheduler (the learner's very first session), the opener is omitted and the topic begins immediately — unseen

material is never used to manufacture an opener; the never-empty rule applies from the first subsequent session onward.

Single-serve rule. An item simultaneously in remediation and due appears **once**, in whichever surface arrives sooner; that one grade writes to both engines.

Comeback protocol. No overdue count is ever surfaced anywhere. Backlog triage = module drag, then R ascending within module. Daily capacity = the opener + at most one "Keeping fresh" and one "Fixing gaps" quiz ready per day; overflow drains across following days; nothing expires or resets. A successful late review yields a larger Stability — the comeback itself deepens memory.

"Keeping fresh" batches. Per module at ≥ 8 due items or the oldest > 3 days past due; size 8–15 (split across ready-days above 15); the due items themselves, interleaved across objectives, no padding; surfaced "Ready today"; standard quiz mechanics.

Interval hygiene. $\pm 5\%$ deterministic per-item fuzz (load spreading) · floor **1 day** — no intraday scheduling; the quiz's wrong-answer lap is the same-day relearning step, and later same-day grades update state without scheduling a same-day return · ceiling = the B2 horizon.

Governance. The scheduler is **FSRS, adopted unmodified**. Launch on FSRS's published default parameters — the median fit across ~20,000 real learner collections, benchmarked on hundreds of millions of logged reviews — with the FSRS release version pinned in SchedulerConfig; the Difficulty/Stability update equations are FSRS's published ones, not restated here. Binary right/wrong grading maps as correct = Good, incorrect = Again and is slightly *more* accurate for this grading style than mixed grading; FSRS natively handles early (not-yet-due) reviews, which the never-empty opener relies on. Log every prediction-vs-outcome; track **cohort RMSE(bins)** — the binned average gap between predicted and observed recall probability — as the scheduler-health KPI. **Per-learner parameter optimization is not built**; build it only if cohort RMSE sustains above 0.07 for a month or a segment shows systematic misprediction (*trigger values: config*).

QA acceptance: never-empty opener; no-same-day-repeat; capacity caps; fuzz determinism; the 1-day floor; horizon ceiling; the no-overdue-count rule verified on every surface.

B5. Session & Flow Composition

In plain terms — the choreographer. It decides what today looks like so the learner never has to: the opener first, the topic's build-time rhythm next, an activity where the content warrants one — and it keeps the rhythm deliberately unpredictable so week nine feels as fresh as week one. Builds on: the templates in the graph (B10) and both engines' queues. Feeds: every session a learner ever has.

Responsibility. Compose every session and every topic's rhythm. Owner of: the flow-template assignments, activity-slot resolution, slot-fill, resume behaviour.

Templates. Each topic's flow is assembled at content-build time from a library of ~8 named templates varying theory-chunk count, quiz position (early / mid / end / split-in-two), and whether an activity slot exists. Constraints: consecutive topics never share a template; any rolling 5-topic window contains ≥ 4 distinct templates and ≥ 1 non-end quiz. Learners never see

template names. **The topic lifecycle across days** comprises six modes — learn · check · explain · recall · prove · revisit — applied at topic level; a single session samples one to three of these modes according to its flow template and engine state.

Activity-slot resolution. Objectives are tagged fact-heavy or conceptual (human-reviewed). Majority fact-heavy → blurring; majority conceptual → Teach Oggi; balanced → whichever the learner has done less recently. **The composer never places blurring and Teach Oggi on the same topic in the same session** (they compete for the same cognitive work in one sitting). This restriction applies to composed sessions only — learner-initiated practice is unrestricted.

Slot-fill (in-flow quiz). Uncovered new items first — **every new item is retrieved at least once before its topic completes, across sittings**; coverage of new items outranks the quiz's length target — then R-ascending; renderings per B1's rules; untimed by construction; grades write to both engines. Topic sizing (A1) makes this fit the session; oversized syllabus content is a build-time error, never a runtime problem.

Routing. Topic completion always routes to the next topic in syllabus order — decision-free; no detour prompts or cross-area suggestions at the completion moment; free browsing everywhere remains. Weakness weighting operates only inside surfaces (opener ranking, slot-fill); the Biggest Opportunity lives on Progress; its actionable work surfaces only in Practice — Recommended.

The recall day. Once per module, the session after its final topic: a blurt on the module's lowest-mean-R objective cluster → a recap of only the blurt-missed chunks → a variant quiz from the misses plus due and remediation items. Standard length and litres; skippable in one tap ("straight on to [next module]?"); skipped material stays with the engines.

Resume. Every re-entry runs the opener first, then resumes at the exact interrupted slot: quizzes at their question index with earlier answers locked; theory at the last page; **an interrupted blurt restarts fresh** (nothing was graded); an interrupted Teach session resolves through End Session. The composer honours a one-shot Debrief priority flag (B11) when one exists.

Session-scoped laws: no time pressure on fresh material; no heavy new content immediately after a mock.

QA acceptance: template-rotation constraints; the pairing rule's scope; the retrieve-before-complete guarantee; resume states; recall-day skippability.

B6. Blurt Scoring

In plain terms — the fair marker for "say everything you remember." The truth is fixed at build time as a set of human-reviewed key points; the marker only ever matches the learner's words against it, with benefit of the doubt as law — so two learners saying the same thing always score the same, and knowing more than the syllabus can never hurt. Builds on: the rubric in the graph (B10). Feeds: both engines, once, at scoring. The same rubric powers B7.

Responsibility. Mark free recall fairly, repeatably, auditably. Owner of: the key-point rubric (shared with B7), blurt results, their single write moment.

The rubric. At content-build time every topic's outcomes decompose into **key points**: atomic, one idea each, ~8–20 per topic, each mapped through the mapping table and tagged **critical** or **supporting**; human-reviewed. The marker only matches learner statements against this fixed rubric.

Matching and statuses. The transcript (typed, or voice-transcribed) is segmented and semantically matched — meaning counts, wording never does. Every key point ends **Recalled / Missed / Incorrect**. Benefit-of-the-doubt law: garbled or unclear statements match nothing; only clear wrongness earns Incorrect. Hint-cued recall counts fully as Recalled. Correct statements outside the rubric are praised and excluded from every denominator — extra knowledge can never lower a score.

The numbers. Coverage = $\text{Recalled} \div \text{total scoped key points}$. Accuracy = $\text{Recalled} \div (\text{Recalled} + \text{Incorrect})$; with nothing included it displays "—", never 0. Critical gaps = the critical-tagged points at Missed or Incorrect, each named in ≤ 10 plain words. Whole percentages only.

The single write (at scoring, nowhere else). $\text{Recalled} - \text{Good per mapped item} \cdot \text{Incorrect} - \text{Again on the specific items} (- \text{remediation}) \cdot \text{Missed} - \text{due-date pull-forward only, no grade}$. The Correct Mistakes flow writes nothing (rewriting with the correction visible is elaboration, not retrieval). The promised follow-up quiz arrives through the standard engines — Incorrect via gap batching, Missed via pull-forward.

Hints. Every key point carries one pre-authored **cue question** (points at the idea, contains none of it). The button cycles the cue questions of not-yet-matched points — critical first, then blueprint weight — one per press; covered points drop out live; exhausted cycles restart through what remains. Deterministic, zero runtime generation, writes nothing.

Voice. One pipeline: transcribe — strip disfluencies (fillers, restarts, repetition) — resolve self-corrections to the learner's **final** version ("wait, no — it's 90 litres" scores the correction) — score identically. Unclear audio matches nothing and can never manufacture an Incorrect. The transcript is what Correct Mistakes annotates.

Annotation. Green = a statement carrying a Recalled point; amber = correct-but-incomplete (Recalled if the core idea is present; the remainder rides the Missed machinery); red = a statement carrying an Incorrect. Jump-to-mistake iterates reds only, in order; every correction skippable; amber never requires correction.

Scope. In-flow blurt = the topic's full key-point set, narrowed to already-covered chunks when the template places it early — never scored against unseen pages. Recall-day blurts scope to their cluster. Practice blurts follow the same rule (full set for completed topics; covered chunks otherwise). Scope fixes at start; Coverage's denominator is exactly the scoped set.

QA acceptance (the simplicity law). The results screen: exactly three numbers and three plain lists; zero system vocabulary; gaps in ≤ 10 plain words; copy at FK grade 6–8; a first-time user can state right / missed / next within 30 seconds in usability testing. Plus: the single-write rule; the ambiguity law; hint neutrality. Any build failing one of these checks fails the feature.

B7. Teach Oggi

In plain terms — the same fair marker wearing a student costume: the protégé effect made measurable. Oggi plays a genuinely confused student who can only know what he's been taught this session; the learner's explanations match against B6's rubric, chips carry the grading so his voice never has to say "wrong," and "Oggi understands it now" is literally a data structure reaching its threshold. Builds on: B6's rubric verbatim. Feeds: both engines, once, at close.

Responsibility. Turn explaining into measured evidence. Owner of: session state (taught set, rung of conversation), the meter, the report, its single write moment. Shares B6's rubric verbatim — one review, one QA surface, directly comparable evidence.

Oggi's knowledge = the taught set. The session's scope is the topic's key-point set (B6's scope rules); "what Oggi understands" is exactly the points successfully taught this session.

Persona constraints (enforced at generation). (a) **The knowledge ratchet:** Oggi's utterances may only contain content from already-taught points — he can restate and attempt what you taught, never introduce untaught material. (b) His questions are the cue questions of unmatched points, rephrased in student voice — deterministic in substance, generative only in phrasing. (c) A flawed explanation earns an innocent follow-up aimed exactly at the flaw ("wait — so it leaks *faster* when he's lower?"), never a correction. (d) The voice never contains "incorrect / wrong / no, that's not right" — chips carry all grading. (e) Asked for the answer, he deflects in character and points at the notes ("ooh — check your notes!").

Chips (per turn). Each explanation is rubric-matched like a blurt statement; multiple points can match in one turn. Great = ≥ 1 point fully and correctly matched · Good = a core idea present but thin · Inaccurate = a core idea with a flawed element · Mistake = a clearly wrong assertion. The ambiguity law holds: unclear teaching matches nothing and can never earn a bad chip.

Meter and panel. A point is **taught** at final status Great/Good. An objective ticks when all its critical points and $\geq \frac{2}{3}$ of its supporting points are taught. Meter = taught weight $\div$ scoped weight, critical points counting double. **Confident** = every scoped critical taught $\wedge$ meter $\geq 80\%$ — the close: thank-you — identity beat — report.

Session flow. Opens confused on the highest-weight critical's gap ("I know a firewall blocks stuff... but I don't get how it decides. Can you teach me?"). Loop: explain — match — chip — next move. Next question = the highest-priority unmatched cue (critical first, then blueprint weight), except flaw-first immediately after Inaccurate/Mistake. If scope is exhausted without Confident, the session closes warmly with the report — never a loop, never a gate.

Teaching Notes. A **static crib sheet**, generated once at content-build time, identical every time it is opened: the scoped objectives in teaching order, and under each, every key point's plain name + cue question — past and future all present from the first minute (glancing anywhere may jog memory). Nothing ever appears, reorders, ticks or highlights; it contains names and questions only, never explanation text (structurally: none exists in its source data); opening writes nothing. Progress signalling lives solely in the objectives panel.

Final-status law and the single write (at close). Within a session each point's final status wins — an early Mistake re-taught correctly ends taught. At natural close or End Session, one

commit: taught → Good per item · final Inaccurate/Mistake → Again on the specific items · never-addressed → pull-forward only. Both close paths also pay litres per objective ticked and produce the report (full on natural close: strengths grouped by objective quoting the learner's best clean formulation; improvements with what was off in plain words; "not covered yet" neutrally; light on End Session: strengths + one top improvement). Transcript attached; when any point ended badly: "recommended learning will pick these up in Practice."

Availability. In-flow via the activity slot (covered-chunk scope) and anytime from Practice → Activities (full-set scope; untaught points simply surface as questions and end at worst "not covered yet"). One live session per topic; re-teaching later always allowed and runs fresh; engines accumulate across sessions. Voice rides B6's pipeline unchanged.

QA acceptance: the ratchet (no untaught content in any Oggi utterance); chip-derivation determinism; the Confident formula; notes immutability and content constraints; the single-write commit identical on both close paths.

B8. The Litre Economy

In plain terms — the price list for the product's one currency. Every learning act pays by what it's worth to memory: first-ever retrieval pays well, retrieval that was genuinely due pays best, re-grinding what isn't due pays almost nothing — which is the entire anti-farm defence, with no locks anywhere. Builds on: scheduler state at the moment of each answer (B4) and the rubric outcomes (B6/B7). Feeds: the flight (B9), with every pump itemised by source.

Responsibility. Price every learning act. Owner of: litre events (source-itemised), the payout table. All numbers below are named constants in **LitreConfig**; no learner-facing knobs; changes never retroactive.

Payout classes (every graded answer, every surface). Class A — first-ever correct retrieval: **5L**. Class B — correct on a due item ($R \leq 0.90$ at serve): **7L** — deliberately the top per-answer rate: maintaining memory outearns meeting it. Class C — correct on a not-due item: **1L** — the single rule that makes re-grinding pointless everywhere (the only anti-farm mechanism; no access restrictions exist). Incorrect: **0L**, and litres never subtract. Remediation serves price as Class B.

Unit rates. Theory page 1L · blurt 4L per Recalled point (hint-cued identical; Missed/Incorrect 0) · Teach Oggi 12L per objective ticked + 10L Confident bonus · Debrief 15L at completion (early exit pays 0 for the Debrief) · topic completion 15L base · recall day +10L.

Bonuses (quiz contexts only — never exam mode, never openers). Hard "big one" questions (~1 in 10, content-flagged) pay 2× their class rate with a beat of announcement drama (copy: "hard question coming — it's the big one"). Combos: +2L on every third consecutive correct; the chain survives across the quiz until a miss; a hard question mid-chain pays its doubled value and the chain bonus. Rare multipliers (granted on a hard-question correct): the remaining questions of that quiz pay 2×/3× class rates — quiz-scoped, never session-scoped. Hard ceiling: **+30L of bonus per session**; past the cap the drama still plays and the counter shows base rates — the presentation never lies about totals.

Assessment payouts. Exam-format runs accrue class rates silently and pay at the feedback screen, plus completion premiums: Exam Simulation +50L · mini-mock +20L · custom exam run +10L. **The premium pays only when $\geq 70\%$ of questions received an answer** (PREMIUM_MIN_ANSWERED); below that, grading still pays the class litres for what was answered. No bonuses of any kind in exam mode. A typical 30-question mock lands ~150–200L in one drop — structurally the largest payment in the product. Pause for Later pays nothing until graded.

The grounded-liftoff guarantee. If a session **started grounded** and **ends with topic completion**, the completion payout becomes **max(15, 100 – session litres earned)**, **capped at 60L**: every honest full session from the ground flies; the cap preserves "the bonus alone never approximates liftoff" (a near-zero-success sitting lands with the gauge nearly full, one opener from the sky). Surfaced as: *"Topic complete — that's liftoff."* Airborne sessions and Save-and-Exit partials always receive the flat 15. **Stated precisely**: liftoff is guaranteed whenever the session earned $\geq 40L$ before the completion payout — which the acceptance simulation validates for every profile with success rate $\geq 55\%$; the surface promise is calibrated to this and is never stated as unconditional.

Reference arithmetic (calibration anchors). Typical quiz-day $\approx 109L$ · Teach-day $\approx 100L$ · blurt-day $\approx 121L$ · a struggling $\sim 55\%$ -success day $\approx 60L$, topped to 100 by the guarantee.

Acceptance (pre-launch simulation over synthetic profiles, success 50–95%, all templates; any failed assertion blocks that LitreConfig): ordinary full sessions land 90–140L (10th–90th percentile) · quiz-, teach- and blurt-days pay within $\pm 20\%$ of each other (no dominant strategy) · grounded + completed $\Rightarrow$ liftoff at any success rate $\geq 55\%$ · the bonus cap binds in $< 5\%$ of sessions · mocks land 150–200L.

B9. Flight Physics & the Event Model

In plain terms — the physics and the diary. Altitude is a pure mathematical function of litres and time — no background jobs, no stored displays, so a stale number cannot exist — and history is an append-only log whose corrections can only ever lengthen a flight, because late-arriving evidence is always extra study. This is the streak problem solved in mechanics rather than copy. Builds on: source-itemised litre events (B8). Feeds: every homepage, the Flight Log, records, awards, the one push, and Wingman surfaces.

Responsibility. Compute altitude, transitions, records, the push, and the honest history. Owner of: the flight event log (the source of truth), flight state cache, records, the Orbit latch.

Constants (PhysicsConfig; tuning ranges in brackets). 1L = 1ft · liftoff 100L · leak below 200ft: 90L/day linear [80–95] · leak at/above 200ft: 50% of current fill/day [40–60%] · proportional threshold 200ft [200–250] · touchdown at fill 0 · ground is leak-free · **cap 1000ft** · push threshold 75L · push idle condition 18h · quiet hours 22:00–08:00 local.

The evaluator (the only place decay maths exists). A pure function evaluate(fill₀, to, t): at/above 200, $F = \text{fill}_0 \cdot 0.5^{(\Delta t/24h)}$; below 200, linear at 90L/day; the 200-boundary crossing solved exactly ($t_c = 24h \cdot \log_2(\text{fill}_0/200)$) with piecewise evaluation across it; exact inverse solves for any threshold (0 → touchdown instant; 100 → episode boundaries; award altitudes —

earned instants). Constants are read from the PhysicsConfig version in force for the era evaluated.

The event log (append-only; the scope rule: record exactly what learner surfaces, Wingman surfaces and admin metrics need — nothing else). Six primary domain event types, plus one correction event type (**annul**, defined below): **pump** (litres, per-source itemisation, idempotent event ID; written at completion beats and every exit) · **liftoff** (written synchronously in the same transaction as its crossing pump; fill = the accumulated ground total G at the crossing, including this pump, so overshoot carries — a first session totalling 109L lifts off at 109ft) · **touchdown** (materialised once at first discovery by any read or write: an atomic conditional append under a one-per-flight unique constraint — racing devices no-op and reload; effective timestamp = the exactly-solved crossing instant, whoever discovers it) · **award-earned** (same materialise-at-discovery pattern, one per award, effective timestamp = the solved crossing) · **notification-sent** (so episode state rebuilds from the log alone) · **config/timezone version markers.** Ordinary reads append nothing — no evaluation records exist; state (fill, mode, running peak, episode state, last event time) is a cache rebuilt deterministically from the log; every displayed value is computed at render from cache + now, so no stale number can exist and login always shows the honest altitude with zero background jobs.

Bi-temporal replay and the correction rule. Every event carries an effective timestamp (physics time) and an append sequence (record time). Physics replays by effective time; append order breaks ties and resolves races. A late-arriving offline pump inserts at its true effective time and forces forward recompute. When late evidence falsifies a materialised transition (a device recorded touchdown at 20:30; another then syncs a 19:00 pump that kept Oggi flying), a **compensating annul event** is appended — the log's seventh, correction-class event type: it references the invalidated materialised event, never deletes it, and participates in replay by effective timestamp and append sequence — state rebuilds, and dependent awards/records re-derive. **By construction, corrections only ever extend flights** (late evidence is always extra litres): "your offline session kept you flying."

Hysteresis. Grounded litres accumulate leak-free; fill $\geq 100 \rightarrow$ liftoff; once airborne, no threshold re-check — Oggi flies until exactly 0. Fill = 0 is touchdown; grounded = 100 exactly is liftoff. Airborne fill hard-caps at 1000; overflow credits lifetime litres only.

Records. Per flight: duration, peak (the running max recorded at each pump — between pumps altitude only falls), litres pumped. Board: Longest Flight · Highest Altitude · Biggest Day (litres per **local calendar day**, stored timezone — the only place local days exist; timezone changes apply forward only) · lifetime litres · lifetime days aloft (exact airborne integral). "Most flights" is never computed, stored or surfaced.

The push. At every event, solve the future instant fill crosses 75L; schedule one candidate at $\max(\text{that instant, last activity} + 18\text{h})$, shifted into the learner's historical study hour, never 22:00–08:00. At fire time, re-validate: any study since scheduling $\rightarrow$ cancel; the episode's allowance already used $\rightarrow$ cancel; else send. **An episode is a continuous period below 100ft; the allowance resets only when fill climbs back above 100** — a small top-up at 60ft clears nothing. **Airborne descents only:** grounded learners are never pushed; the ground is a peaceful state.

Display states (pure render function). Airborne ≥ 100 → exact live altitude. Airborne < 100 → the same honest altitude + the single coming-in-low line ("I'm drifting low. One lesson tops me up") — the only place the interface comments on decline. Grounded → the takeoff bar with the exact figure; no decay shown because none occurs. Grounded at 0 with prior flights → the warm post-touchdown story on next open.

The Extended Sky. Twelve bands of realistic scenery — **the following inventory is art-direction law:** 0ft roads, pavements, ground-level building walls, entrances, parked vehicles, streetlights and drains · 100ft low-rise rooftops, houses, shops, small offices, trees and billboards · 200ft mid-rise apartment blocks, warehouses, parking structures and larger signs · 300ft taller apartment buildings, office blocks, rooftop water tanks and antennas · 400ft high-rise buildings begin to dominate; streets become less visible · 500ft skyscraper rooftops, cranes, radio masts, rooftop machinery and helipads (**the Skyline**) · 600ft mostly upper skyscraper sections, construction cranes and communication towers · 700ft only the tallest towers, spires, aerials and distant skyline shapes · 800ft a few landmark skyscrapers and very tall broadcast towers · 900ft tower tips, aircraft warning lights, clouds and open sky · 1000ft mostly sky and clouds · beyond: Orbit. Bands carry no learner-facing names (copy speaks in feet); names exist on badges only (A14). Pace ladder: cruise $\approx 2\times$ daily litres; $\sim 250\text{L/day}$ breaks the Skyline; $\sim 500\text{L/day}$ sustains 1000. **Orbit:** a latched display state on any cap-overflow pump — the altitude readout shows "?? ft" (the one deliberate exception to exact-figures; the Helium readout stays exact and the log/records always carry the honest 1000), the scene shows Oggi in orbit — **the city far below, Earth's curve and stars** — and it unlatches when fill decays below 900 ($\sim 3\text{--}4$ hours per overflow). **The cap-overflow pump's climb animation plays the orbit-break beat.** Copy: "I can see the whole city from here. Actually... I can see everything."

Edge cases. Same-day relaunch is a new flight; Biggest Day sums across flights per local day. Multiple pumps evaluate piecewise at their own timestamps; out-of-order sync inserts at true time with forward recompute (the one mandatory cache rebuild). Overflow at the cap records peak = 1000, never more.

Analytics (from events alone). Touchdown frequency = touchdowns per learner per 28 days among weekly-active learners · weekend-cohort grounding rate = the share of learners whose pumps concentrate $\geq 70\%$ on Saturday/Sunday who experience ≥ 1 touchdown per 28 days (the cohort the proportional-leak zone protects) · **D7-after-touchdown** = the share of touchdowns followed by any pump within 7 days (the north-star forgiveness metric) · the push funnel from notification-sent events. **The causation rule:** config versioning yields clean attribution, never causation — physics settings change only after sufficiently large cohort comparisons or controlled experiments.

QA acceptance: two independent implementations of the evaluator produce identical altitudes; reads-append-nothing; atomic single materialisation under device races; annul-only corrections that never shorten a flight; the episode-reset rule; version stamps on every event.

B10. The Content Pipeline & Item Graph

In plain terms — the factory and the single source of truth. One versioned graph holds everything learners ever see and everything the systems ever read — and nothing publishes without passing its build-time gates and its human review. Every rubric, cue question, rendering,

Teaching Note and confusable edge in this specification is a view of this graph. Feeds: literally everything; nothing else may originate content.

Responsibility. Produce and maintain everything learners see. Owner of: the versioned content graph (the single content source of truth — every runtime structure is a view of it).

The graph. Qualification — Modules (blueprint weights) — Topics (template, sizing) — Objectives (fact/conceptual tags, sub-weights) — Key Points (critical/supporting, plain names, cue questions) — Knowledge Items — Renderings (base + ≥ 2 variants + ladder versions, hard flags), with one-idea lesson chunks linked to objectives and key points. **Every constant in this specification is a build-time validation gate** (5–10 items/topic · 8–20 key points/topic · ≥ 3 renderings/item · ~1-in-10 hard flags · template rotation rules · the authoring law of A3): a topic failing any gate cannot publish.

The chain. Blueprint ingestion (official syllabus — objectives + weights) — decomposition (— key points, cue questions, tags) — drafting (lesson pages under A3, base questions, worked examples) — derived manufacture (variants, ladder versions, worked-example fade sequences, hard flags, static Teaching Notes, interleave links) — **the human review gate** — publish.

Every learner-facing artifact is human-approved before going live — including manufactured variants (reviewed batched beside their base) and any regeneration triggered by exhaustion events, which re-enters the gate. Automated pre-checks (answer preservation, no new facts, reading level, word caps) reduce reviewer load and never replace approval. **Artifact boundary:** an "artifact" is any pre-authored content unit (pages, renderings, notes, cue questions, recap chunks, worked examples). Runtime AI utterances are not artifacts; they are constrained instead by the grounding law and role contracts (B11) to gate-approved source facts — a fresh example generated in conversation must derive strictly from the item's approved key point and worked-example data and may never introduce new syllabus claims.

Question-bank imports. Uploaded banks enter at the drafting stage: each question is auto-mapped to its objective and key point; **near-duplicate detection** runs against the target item's renderings and the module bank (duplicates merge or discard, never stack); the question attaches as a new rendering on an existing item (counting toward the ≥ 3 minimum) or seeds a new item; reviewers must **confirm the mapping itself**, not merely the wording (a well-written question mapped to the wrong objective corrupts mastery, readiness and spacing simultaneously); imports feed derived manufacture like drafted content. **No provenance or licence data is stored** — rights clearance is a wholly out-of-band team responsibility (risk recorded in Document 4).

Confusable edges. Human-confirmed "confusable with" links between items/objectives are the **only** source for interleaved composition and gap-quiz padding; edges exist only within a module or across explicitly related modules; **no edge, no mixing**. Edge coverage is reported per module.

QA loop. One inbox; weekly human triage; dispositions: fix (a corrected rendering; scheduler state carries) · add variants · retire. Active launch signals: learner error reports + one statistical tripwire (cohort first-attempt success $< 30\%$, $n \geq 30$). Concept-breakdown and inventory-exhaustion events log continuously and surface in the weekly batch. **The urgent route:** an "urgent — seriously wrong or harmful" category on the learner report sheet, plus staff flagging — immediate notification, a one-click pause (the item stops serving instantly and is excluded

from new scheduling; mastery denominators untouched pending disposition), **auto-pause at ≥ 2 independent urgent reports** (one alone never auto-pauses), and a **48-hour fix-or-retire SLA**. Escalation to a full severity-ranked queue is pre-defined: sustained report volume or anomaly rate above threshold for a month (*threshold values: Open Decisions*).

Retirement. A fix preserving *what is tested* ships as a corrected rendering on the same item. A change to what is tested **retires** the item: it stops serving immediately, **drops out of every mastery and readiness denominator**, and archives its scheduler history; replacements enter as new items at zero.

Editions. Each awarding-body syllabus edition is a graph version; learners bind at enrolment; analytics events carry the graph version. Updates diff into: unchanged (carry) · modified-same-knowledge (corrected renderings) · changed-knowledge (retire + new) · added/removed topics (recompose; retire). Existing learners stay on their edition until moved by explicit team policy — never a surprise reset.

QA acceptance: gate enforcement (unpublishable failures); the human-approval invariant; import dedupe + mapping confirmation; the pause SLA; denominator behaviour on retirement.

B11. Oggi's Underlying Systems

In plain terms — Oggi's brain-stem: exactly what he is given (a deterministic context frame, never raw access), exactly what he remembers (the engine states themselves plus a small, fully inspectable notes store — so his memory can never disagree with mastery), and exactly what he may say (role contracts, a three-rung ladder, a hard capability boundary). His initiative never exceeds an invitation. Builds on: the graph and the engine states. Feeds: every conversational surface in Part A.

Responsibility. Ground, bound and remember. Owner of: the context frame, the notes store, role contracts, the hint ladder, unstick offers, prompt programs, the Debrief mechanics, the capability boundary.

The context frame. Every invocation receives a structured frame, never raw access: page context (current chunk/rendering + the learner's answer + the correct answer + explanation) · knowledge context (the item's key point, objective, confusable edges, distractor-rationale data) · learner context (mastery snapshot, recent remediation flags, session position) · the role contract. **Grounding law:** substantive claims must trace to the frame; beyond it (general chat aside), Oggi says so plainly rather than improvising syllabus facts. Frames are logged per invocation for reproducible QA.

Memory: two stores. (a) **The engine states are the learner model** — every item's R, Stability and remediation flag; no separate knowledge diary exists, so Oggi's picture can never disagree with mastery. (b) **A small notes store:** derived notes (auto-generated from engine data — "keeps confusing X with Y" — regenerated as evidence changes) and told notes (facts the learner explicitly shares that matter pedagogically, stored **only with an in-chat acknowledgement**: "noted — I'll remember that"). The transparency panel renders store (b) verbatim, every entry editable/deletable, derived entries labelled. **A sensitive-topics filter blocks told-note writes entirely** — nothing sensitive can be stored, silently or otherwise.

Support transcripts: short-term retention for the report button and QA only, never mined into context — which is why chats look fresh while Oggi still knows you.

The greeting. One line maximum; input free from the first moment. With a genuinely storied item (remediation history or a long gap on a past struggle), the greet may lead with an **invitation carrying a visible alternative** ("Want to pick back up where subnetting got tricky — or something else?"); one tap serves that item as the opener's first question — one serve, graded and paid normally; ignoring costs nothing and it is not re-asked that session; maximum one offer per session; with nothing storied, plain warmth — manufactured callbacks are banned.

The hint ladder (per question instance). Three rungs: **hint** — **worked step** — **answer**. Advance only on explicit learner request, exactly one rung per request; "just tell me the answer" from cold advances to the hint, in character ("let's get you there — try this first"); the answer is reachable only through the worked step; the ladder resets per question instance (a variant days later starts fresh); rung content derives from the key point/cue question and the worked-example artifacts — grounded, never improvised. The quiz's 50/50 button is a separate fixed control. Exam mode has no ladder — Oggi is absent.

Role contracts (every invocation carries one). **Tutor:** full frame access; ladder discipline; Socratic default — questions and prompts outrank explanations unless explanation is explicitly requested or the ladder has reached it (the 80%-learner-words principle operationalised). **Student:** the Teach persona constraint set (B7). **Facilitator:** the Debrief — may reference the module's covered material, introduces no new content, produces questions and connections only. Global laws bind all three: the banned vocabulary; first-person flight copy; the capability boundary. **Voice authority: the team's tone-of-voice documents, one per rule;** the enforcement layer's banned-pattern list derives from them. Enforcement is two-layer: contract-shaped generation plus an output screen — a violation regenerates once, then falls back to the safe scripted line. Outputs are tagged with their contract version.

The Debrief (five computable stages; Facilitator contract). Scope = the module's full key-point set + its confusable edges. 1) The learner's two-sentence summary is rubric-matched against the module's highest-weight critical points. 2) Two–three probes from the cue questions of unmatched high-weight points, phrased through why/what-if templates. 3) One connection prompt from cross-module edges (skips gracefully when none exist). 4) The learner names the shakiest area — mapped to its nearest objective — a due-date pull-forward on that objective's items **plus a one-shot priority flag the next session's composer consumes** — how "feeds tomorrow's plan" is mechanically true. 5) A preview of the next module read from the graph — no new teaching. Writes are light-touch under the ambiguity law: clean matches — Good; clear wrongness — Again; unaddressed — nothing. 15L at completion; early exit pays zero; Practice re-runs identical, module-scoped.

The unstick offer (OggiConfig). Triggers: a 2nd consecutive incorrect in a quiz · 90 seconds stalled on a visible unanswered question (paused when backgrounded) · a repeat miss in the wrong-answer lap (reteach offer). Caps: once per question instance; ≤2 offers per session; dismissal **or non-engagement** sets a never-again flag on that instance. Accepting opens Tutor-Oggi with the wrong-answer prompts preloaded. Structurally never fires in exam mode, blurts,

Teach, or theory reading (lingering is thinking). The offer is one ignorable line: "Want a hand with this one?" Offer/accept/dismiss events are logged for threshold tuning.

Quick prompts as programs. Every prompt ID maps to: context requirements (unmet → the prompt does not appear — how "max four, always contextual" resolves) · a grounded action bound to graph data ("Reteach with a new example" → the item's alternate worked example or one generated strictly from the key point; "What's the common trap here?" → distractor-rationale data; "How would the exam ask this?" → the objective's exam-format data) · its role contract. Typed questions run under the same contract and grounding law. Every resolution logs prompt ID + frame version.

The capability boundary. Two fixed deflections via the same lightweight intent routing that selects contracts — no classifier network, no resource tables, no therapising, no check-in machinery: platform/account matters → a warm handoff to the support desk ("that's one for the Ogwi support team — here's how to reach them"); personal, emotional, medical or legal territory → one fixed, polite tone-of-voice line that he doesn't have the capabilities for that, said warmly and without judgement, with no advice clause, remaining available for study. The sensitive-notes filter stands; **the conversation never ends on Oggi's side**; nothing is ever reported to anyone.

QA acceptance: grounding traceability on sampled frames; ratchet and vocabulary screens; ladder transitions; single-offer greet; note-write acknowledgement; sensitive-write blocking; boundary deflections; contract-version tagging.

B12. Cross-System Semantics

In plain terms — the glue: the exact meanings that make the sustained awards computable from the flight log, plus the pointer into Part C, where the map, the invariants and the governance close the whole machine. Builds on: B9's event log. Feeds: the awards surface (A14) and the release gate.

Sustained-award semantics. Residency — daily peak $\geq$ threshold on N consecutive local calendar days — governs Tower Residency (400×7), Skyline Week (500×7), Skyline Fortnight (500×14), Spire Week (700×7); binge-proof (one 700L Sunday buys ~2 qualifying days, never 7); computed from per-local-day running maxima, which occur only at pumps. **Continuous-hold** — fill never below threshold across a continuous 24 hours — governs **Spire Day** (700); troughs occur only just-before-pumps, so the check is the evaluator at pump instants; genuinely achievable only because the sky reaches 1000 (~600L/day with top-ups). **Cruise Control** — time-weighted mean altitude ≥ 300 across an entire 30-day flight — the exact integral of the piecewise decay between pumps. All discover via award-earned materialisation with exactly-solved timestamps.

The canonical map, event model, configuration registry, publish points, invariants and sources of truth are reproduced in Part C below and are normative in this document; the closure rule — no data-flow arrow may exist in code that does not exist on the map — is release-blocking.

PART C — ARCHITECTURE & GOVERNANCE (NORMATIVE)

You have now read every behaviour. Part C is the skeleton those behaviours hang on — who owns what, what may flow where, when things recompute, which numbers may ever be tuned, and the twelve guarantees that block any release. Read it before writing code; return to it before adding any new data flow (the closure rule below makes that mandatory, not advisory).

C1. Canonical entity & data-flow map

The hub: the mapping table (owned by B1, authored by B10).

CONTENT GRAPH (B10) —views→ mapping table · rubrics · cue banks ·
Teaching Notes ·
renderings · lesson chunks ·
confusable edges · templates

RUNTIME STATE	owner
per-item memory states (D,S,R)	B4 (read by B1, B2)
remediation bank + rungs	B3
flight event log	B9
notes store (Oggi)	B11
published mastery/odds/forecast	B1 / B2

FLows

graded answer (any surface) → one review event → B4 state update
AND B3 entry/exit
blur / Teach / Debrief scoring → per-key-point writes → same
review-event path
B4 due sets + B3 batches → openers & Recommended (composed by B5)
litre events (B8, source-itemised) → B9 pumps → altitude,
records, awards
forecast (B2) → spacing horizon → B4 interval ceilings
engine states + graph → Oggi context frames (B11); Oggi writes
only notes + ordinary grades
published values only → Progress, Wingman, admin surfaces

Closure rule (release-blocking): no data-flow arrow may exist in code that does not exist on this map; adding one requires a map revision first. Nothing reads a raw store except its owning system.

C2. Event-model summary

The flight log is append-only with six primary domain event types — pump (source-itemised, idempotent ID) · liftoff (synchronous with its crossing pump) · touchdown (materialised once at first discovery; atomic one-per-flight; effective timestamp = the solved crossing) · award-earned (same pattern, one per award) · notification-sent · config/timezone markers — plus the **annul** correction type, which references the invalidated event, never deletes it, and participates in replay. Ordinary reads append nothing; displays compute at render from cache + now. Bi-

temporal: effective timestamp for physics, append sequence for ties/races; corrections only ever extend flights. One pure evaluator function owns all decay maths. (Full rules: B9.)

C3. Configuration registry & governance

Config	Owner	Contents (examples)
LitreConfig	B8	class rates 5/7/1 · unit rates · bonus values · +30L session cap · premiums 50/20/10 · PREMIUM_MIN_ANSW ERED 70% · guarantee cap 60
PhysicsConfig	B9	leak 90L/day · 50%/day · threshold 200 · liftoff 100 · cap 1000 · push 75L/18h · quiet hours
SchedulerConfig	B4/B2	desired retention 0.90 · fuzz ±5% · floor 1d · horizon 45d/12mo/3mo default · optimizer trigger RMSE 0.07 · FSRs version pin
MasteryConfig	B1	displayed-decline half-life 7d · daily-rollover timing
OggiConfig	B11	unstick 2-wrong / 90s / lap-repeat · ≤2 offers/session · greet 1/session
ReadinessConfig	B2	14d projection · clamp [0.7,1.1] · σ floor · 20% display floor · 80% celebration · band thresholds
Content-graph versions	B10	one immutable version per syllabus edition

Governance law (all configs): immutable, timestamped, versioned documents; forward-only application; every event stamped with the versions in force; replay selects the version for its era; physics-affecting changes require cohort-scale comparison or controlled experiment before adoption (versioning gives attribution, never causation); no config is learner-facing. Values assigned to these documents are tunable; **every other numeric value in this specification is law.**

C4. Publish-point table

Moment	Recomputes
Every graded answer	item memory state (B4) · remediation entry/exit (B3)
Every pump / exit / read	flight state at render (B9)
Session end / Practice-run end	published mastery (B1) · Biggest Opportunity (B1) · odds + certainty + next action (B2) · forecast (B2)
Daily rollover (inactivity)	the same four, once
Blurt/Teach/Debrief scoring moment	their single engine write
Engine batch-queue changes (B3/B4 thresholds or ready-days)	Recommended's contents (A9 §4)
Never mid-activity	any published display value

The publish-point taxonomy itself — which moments exist, and the never-mid-activity rule — is law; only the daily rollover's clock time is tunable (MasteryConfig).

C5. Cross-system invariants (one always-green suite; any red blocks release)

1. Litres never subtract.
2. Ordinary reads append nothing to the event log.
3. Flight-history corrections only ever extend flights.
4. One serve — both engines written; an item is never double-served.
5. No learner-surfaced prediction is computed from a smoothed input.
6. Mastery can fall only via a failed previously-known item or time decay.
7. Nothing expires, resets, or is a deadline.
8. Every learner-facing artifact passed the human review gate.
9. Every event carries its config-version IDs.
10. Exam mode contains no aid machinery.
11. Sensitive content is structurally unstorable.
12. Attempt counts surface nowhere.

C6. Single source of truth

Value / state	Owner (authoritative)	Everyone else
Content, rubrics, cue questions, notes, edges, templates, weights	the versioned content graph (B10)	views only
item $\leftrightarrow$ objective $\leftrightarrow$ module $\leftrightarrow$ weight	the mapping table (B1)	read-only
Per-item memory (D, S, R)	scheduler state (B4)	derived reads
Remediation membership + rung	the bank (B3)	via its 3-signal interface
Published mastery / states / Biggest Opportunity	B1 publishing	display

Value / state	Owner (authoritative)	Everyone else
Odds, certainty, next action, forecast, horizon	B2	display / B4 ceiling
Litre pricing	LitreConfig via B8	none
Flight history, records, awards, episodes	the event log (B9)	caches/views
Oggi's notes	the notes store (B11)	the panel renders it verbatim
Voice & copy	the Tone-of-Voice appendix (required input; Part D)	enforcement derives from it
Pass mark	the qualification record (from the awarding body)	all thresholds

PART D — OPEN DECISIONS

Everything above is decided; this short list is everything that is not. Each item names who resolves it and when. If a question arises during the build that is not answered by this document and not listed here, it is a new open decision — raise it; never invent the answer.

1. **First vertical:** final qualification selection from the candidate pool against the four stated criteria.
2. **First-mock invitation threshold:** the mastery level that triggers the invitation (a LitreConfig-style constant awaiting calibration).
3. **QA escalation thresholds:** the numeric report-volume/anomaly-rate values that trigger the full severity-ranked queue.
4. **Hard-question selection rubric:** the content-ops criteria by which ~1-in-10 questions earn the hard flag.
5. **The concrete flow-template library:** the ~8 named sequences, authored during content production within B5's constraints.
6. **Required input — the Tone-of-Voice appendix:** the approved tone-of-voice rules, banned-pattern lists and safe fallback scripts must be attached to this handover as a normative appendix before Oggi's voice layer is finalised. The structure, contracts and enforcement mechanics are fully specified (B11); the words themselves are governed by that appendix and cannot be reconstructed from this document set.

Nothing else is open. Any behaviour not specified in this document set must be raised, not invented.